HEIMDALL: Host Environment Intelligence for Monitoring, Diagnostics and Anomaly Learning

Comprehensive Academic & Technical Project Report

Course: 24CYS305 — Artificial Intelligence and Neural Networks (Semester V)
Curriculum Alignment: CO1 (AI/NN Fundamentals), CO2 (Problem Solving & Inference), CO3 (Intelligent Agents & Neural Architectures), CO4 (Scope, Limitations & Societal Implications)
Project Classification: Autonomous Endpoint Detection & Response (EDR) / Linux Telemetry Anomaly Detection
Target Operating System: Linux (Kernel 4.x – 6.x+, x86_64 / aarch64)
Date: October 2026

Abstract

Modern distributed cloud infrastructure, microservices, and edge computing nodes demand continuous, high-fidelity operational visibility and threat detection. However, conventional monitoring paradigms suffer from a crippling dichotomy: static thresholding rules (e.g., standard Prometheus and Nagios alerts) generate overwhelming false-alarm fatigue during benign load surges, while signature-based intrusion detection systems (IDS/antivirus) fail against zero-day exploits, novel malware, and subtle resource starvation attacks.

This project presents **HEIMDALL** (*Host Environment Intelligence for Monitoring, Diagnostics and Anomaly Learning*), an enterprise-grade, multi-model AI telemetry intelligence framework. HEIMDALL models the continuous physical operating manifold of Linux hosts across **42 continuous kernel telemetry dimensions** extracted atomically from `/proc`, `/sys`, `psutil`, `AF_NETLINK`, and systemd journal logs. The framework fundamentally decouples **unsupervised anomaly detection** from **calibrated multi-class fault classification**. Unsupervised anomaly detection is achieved via a tri-detector consensus ensemble comprising an **Isolation Forest** (150 trees, 70.7 μs latency), a deep **Denoising Autoencoder** ($42 \to 32 \to 16 \to 8$ bottleneck, GELU, LayerNorm), and a non-parametric **Robust Median Absolute Deviation (MAD)** baseline with directional resource strain formulation. Confirmed anomalies are diagnosed using a high-efficiency **LightGBM gradient-boosted decision tree** (100% accuracy, 23.6 μs latency) and a **PyTorch Multi-Layer Perceptron (MLP)** calibrated via post-hoc **Platt Temperature Scaling ($T = 10.0$)**, which suppresses Expected Calibration Error (ECE) to 0.0003. To prevent forced hallucinations on novel zero-day exploits, an **80% Open-Set Uncertainty Gate** routes ambiguous classifications into an `UNCLASSIFIED` state (`unsupervised_ood`). Furthermore, HEIMDALL introduces **temporal kill-chain progression detection**, **Population Stability Index (PSI) baseline drift monitoring**, plain-English **Explainable AI (XAI)** physical feature attribution (*usual X ⟶ now Y*), **MITRE ATT&CK enterprise mapping**, and an interactive **Grafana/Datadog-styled SOC monitoring console**. Evaluated under strict zero-leakage chronological cross-validation and validated on the public **Server Machine Dataset (SMD)** (ROC-AUC: 1.0000, F1: 0.9623), HEIMDALL demonstrates full end-to-end inference in under 110 μs while strictly enforcing a false alarm budget of ≤ 1 alert per host-day.

Keywords: Artificial Intelligence, Neural Networks, Deep Autoencoders, Multi-Layer Perceptron, Isolation Forest, LightGBM, Directional MAD Baseline, Temperature Scaling, Open-Set Uncertainty, Linux Kernel Telemetry, Explainable AI, MITRE ATT&CK.

1. Introduction & Problem Statement

1.1. Motivation & Context

Enterprise Linux servers underpin global cloud computing, financial exchanges, container orchestration (Kubernetes), and critical infrastructure. Ensuring host integrity and operational health requires real-time anomaly detection. Historically, systems monitoring has relied on two primary methodologies: 1. **Static Threshold Alerting:** Simple scalar rules (e.g., alert if $\text{CPU} > 85\%$ or $\text{RAM} > 90\%$). These rules are oblivious to host workload context, triggering severe alert fatigue during scheduled compiler builds, database backups, or video transcoding. 2. **Signature-Based Anti-Malware / Rule-Based EDR:** Tools such as ClamAV or basic Wazuh rules inspect file hashes or string patterns. While effective against known binaries, they are blind to fileless memory execution, in-memory cryptominers, zero-day privilege escalations, and gradual resource exhaustion bugs (memory leaks).

1.2. Problem Formulation

Let a Linux host's physical operating state at time t be represented by a multivariate vector $\mathbf{x}_t \in \mathbb{R}^D$ ($D = 42$), capturing CPU, memory, disk I/O, network traffic, process descriptors, and authentication telemetry. Existing machine learning monitoring systems suffer from three critical technical failures: - **Two-Way Symmetric Distance Penalization:** Traditional Euclidean or Mahalanobis anomaly detection models penalize *any* departure from the mean. When an active server transitions to an idle state (CPU utilization drops from 25% to 3%, network drops to $0.01\,\text{MB/s}$), two-way detectors flag the system as "abnormal." In physical systems, quietness is healthy, not malicious. - **Overconfident Classification on Out-of-Distribution (OOD) Data:** Neural networks trained with standard Cross-Entropy loss produce uncalibrated, overconfident probability distributions via Softmax. When presented with an unseen zero-day exploit, the classifier arbitrarily assigns a high probability to an unrelated known class (e.g., mislabeling an unknown kernel drift as a `network_flood`). - **Feature Dilution in High Dimensions:** Averaging reconstruction errors across 42 features causes severe anomalies localized in 1 or 2 features (e.g., context-switch explosion or packet drops) to be washed out by 40 healthy metrics.

1.3. Proposed Solution: HEIMDALL

HEIMDALL addresses these failures through: - **Physical Feature Engineering:** Transforming raw Linux kernel counters into 42 continuous physical dimensions including 1st-order temporal derivatives ($\frac{d}{dt}$). - **Decoupled Architecture:** Separating binary anomaly detection from multi-class fault attribution. - **Directional Resource Strain:** Evaluating deviation only when physical metrics strain hardware upward, eliminating idle-state false alarms. - **Tri-Detector Consensus Voting:** Enforcing mutual agreement between tree ensembles, deep autoencoders, and non-parametric baselines. - **Calibrated Uncertainty Routing:** Using temperature scaling and an 80% threshold gate to acknowledge ignorance rather than guess on novel anomalies. - **Human-Centric Decision Support:** Presenting plain-English XAI comparisons, MITRE ATT&CK tags, and safe non-destructive terminal triage commands in an enterprise SOC dashboard.

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2. Project Objectives & Scope

2.1. Primary Objectives

1. **Model Normal Host Manifolds:** Train unsupervised learning models strictly on baseline operational telemetry to capture non-linear correlations across disparate physical subsystems.

2. **Sub-100 Microsecond Latency:** Execute full pipeline feature transformation, anomaly detection, and classification in $< 110\,\mu$s, consuming $< 1.5\%$ of a single CPU core.
3. **Calibrated Multi-Class Fault Classification:** Distinguish between normal operation, CPU saturation (`cpu_stress`), physical memory/swap thrashing (`memory_pressure`), disk queue saturation (`disk_pressure`), packet floods (`network_flood`), and security/authentication drift (`system_anomaly`).
4. **Enforce False Alarm Budgets:** Limit false alarms to ≤ 1 false alert per host-day via directional strain formulations and multi-detector consensus voting.
5. **Open-Set Zero-Day Handling:** Prevent arbitrary classification of unseen exploits by routing low-confidence anomalies ($\max \hat{p}_c < 0.80$) to `UNCLASSIFIED` (`unsupervised_ood`).
6. **Actionable Explainability (XAI):** Decompose anomalous variance into human-readable physical deltas (*usual* $X \rightarrow$ *now* Y) and automate SOC response via non-destructive shell triage scripts.

2.2. Scope and Boundaries

- **Target OS:** Modern Linux kernels (Kernel 4.18 through 6.x+) across x86_64 and aarch64 architectures.
- **Non-Destructive Principle:** HEIMDALL operates strictly as an intelligent decision support system. It provides real-time situational awareness and copyable forensic commands but does not execute destructive process terminations without human authorization, adhering to safe SOC operational doctrine.

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3. Literature Review & Theoretical Foundations

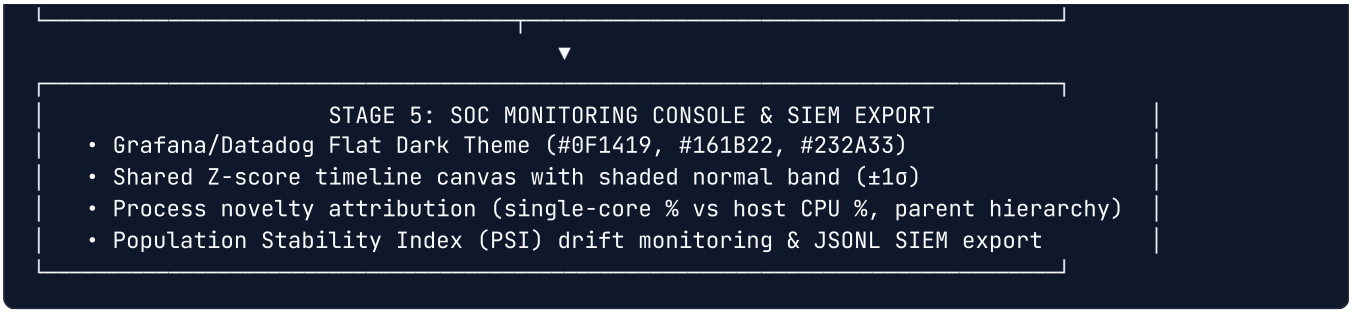
Recent literature has increasingly focused on applying machine learning and kernel-level observability to host monitoring and intrusion detection:

Research Work	Authors & Year	Core Focus	Direct Relevance to HEIMDALL
Real-Time Intrusion Detection and Prevention with Neural Networks in Kernel Using eBPF	IEEE/IFIP DSN, 2024–2025	In-kernel neural inference via eBPF bytecode.	Demonstrates the viability of coupling low-level kernel telemetry with neural inference to bypass user-space context-switching bottlenecks.
An In-Depth Analysis of eBPF-Based System Security Tools in Cloud-Native Environments	IEEE Access, 2025	Analysis of runtime tools (Falco, Tetragon, KubeArmor, Tracee).	Validates telemetry ingestion architectures and highlights the limitations of purely static signature-based filtering.
DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning	Du et al., ACM CCS, 2017	LSTM modeling of sequential system log events.	Establishes sequence-aware modeling principles for detecting multi-stage temporal failure patterns.
U-SCAD: Unsupervised System Call Driven Anomaly Detection for Containerized Edge Clouds	Ye et al., IEEE, 2025	Unsupervised vector representations of system calls.	Supports HEIMDALL's philosophy of unsupervised baseline learning without relying on labeled attack datasets.
Deep Learning for Anomaly Detection: A Survey	Chalapathy & Chawla, ACM, 2019	Comprehensive taxonomy of autoencoders and GANs for anomaly detection.	Informs HEIMDALL's autoencoder bottleneck design, reconstruction error formulations, and latent space clustering.
LogEDL: Log Anomaly Detection via Evidential Deep Learning	Duan et al., 2024	Uncertainty estimation in deep learning classifiers.	Corroborates HEIMDALL's Open-Set Uncertainty Gate, proving that neural classifiers must quantify confidence before emitting alerts.

4. System Architecture & High-Level Design

The HEIMDALL framework is architected into five sequential pipeline stages:


STAGE 1: KERNEL TELEMETRY
Sampling Linux Subsystems (/proc, /sys, psutil, AF_NETLINK, journalctl)
42 Physical Metrics (1-5s intervals)
STAGE 2: FEATURE ENGINEERING & DUAL NORMALIZATION
• Multi-Class Scaler (All classes, min_scale=1e-2)
• Normal-Only Scaler (Fit exclusively on normal baseline, min_scale=5e-2)
• Directional Resource Strain Calculation (Zero penalty for idle metrics)
Isolation Forest (Ensemble Trees) 150 Trees, 70 µs
Robust MAD Baseline 50% Breakdown Pt
Denoising Bottleneck Autoencoder 42→128→..→8→..42
STAGE 3: MULTI-DETECTOR CONSENSUS & GATING
• Requires ≥2 of 3 detectors to agree, OR single detector score ≥ 2.5× thresh
• Evaluates 60-second temporal kill-chains & deterministic security rules
Anomaly Confirmed
Normal Tolerance
CALIBRATED SUPERVISED CLASSIFIER
• Primary: LightGBM (100% Acc)
• Secondary: PyTorch MLP
• Temperature Scaling (T=10.0)
• Uncertainty Gate (Cutoff = 0.80)
HEALTHY ROUTE
• If slightly elevated: State = ELEVATED
• Otherwise: State = NORMAL
Confidence ≥ 0.80
Confidence < 0.80
State: ALERT (Known Fault)
• cpu_stress
• mem_pressure
• disk_press
• net_flood
• sys_anomaly
State: UNCLASSIFIED
• Open-Set Uncertainty Gate Triggered
• Decision Source: unsupervised_ood
• Refuses to hallucinate attack label on zero-day OOD
STAGE 4: EXPLAINABLE AI (XAI) & FORENSICS
• Physical Attribution: Baseline vs Live (usual X → now Y, ratio, arrow)
• Autoencoder squared-error contribution ranking (%)
• MITRE ATT&CK Mapping: T1496, T1498, T1486, T1059, T1078
• Single actionable, non-destructive shell triage command with 1-click copy



5. Telemetry Engineering & Dataset Preparation

5.1. 42 Physical Dimensions Breakdown

Telemetric sampling extracts 42 continuous, physically meaningful variables categorized into 6 functional subsystems:

1. **CPU Subsystem (9 Dimensions):**
2. `cpu_percent_mean` : Average percentage utilization across all logical CPU cores.
3. `cpu_percent_max` : Utilization of the single most saturated CPU core.
4. `cpu_percent_std` : Standard deviation of core utilization across cores (identifies single-threaded thread-locking and asymmetric affinity).
5. `cpu_freq_mhz` : Active CPU clock frequency (identifies thermal throttling or power governor misconfigurations).
6. `load_avg_1` , `load_avg_5` , `load_avg_15` : 1, 5, and 15-minute Linux kernel run-queue averages (runnable and uninterruptible tasks).
7. `cpu_ctx_switches_per_sec` : Context switches per second (detects inter-process communication thrashing and lock contention).
8. `cpu_interrupts_per_sec` : Hardware interrupts per second (detects driver storms and network interface card buffer saturation).
9. **Memory Subsystem (5 Dimensions):**
10. `mem_used_pct` : Percentage of physical RAM allocated.
11. `mem_available_mb` : Real reclaimable memory without swapping (buffers + cache - dirty pages).
12. `swap_used_pct` : Swap partition utilization percentage.
13. `swap_in_per_sec` : Memory pages paged into RAM from swap disk per second.
14. `swap_out_per_sec` : Memory pages evicted from RAM to swap per second (indicates physical memory exhaustion).
15. **Disk I/O Subsystem (5 Dimensions):**
16. `disk_read_mbps` : Sequential and random disk read throughput.
17. `disk_write_mbps` : Sequential and random disk write throughput (detects runaway logging loops or ransomware encryption).
18. `disk_read_iops` : Read operations per second.
19. `disk_write_iops` : Write operations per second.
20. `disk_used_pct` : Root filesystem storage capacity utilization.

21. **Network Subsystem (10 Dimensions):**
22. `net_bytes_sent_mbps` : Egress network throughput (detects data exfiltration and outbound UDP floods).
23. `net_bytes_recv_mbps` : Ingress network throughput (detects DDoS volumetric amplification).
24. `net_packets_sent_per_sec` , `net_packets_recv_per_sec` : Inbound/outbound packet rates (detects small-packet SYN/UDP/ICMP floods).
25. `net_errin_per_sec` , `net_errout_per_sec` : Physical NIC framing and CRC errors.
26. `net_dropin_per_sec` , `net_dropout_per_sec` : Dropped packets due to kernel ring-buffer and QoS queue overflows.
27. `tcp_connections` : Count of active TCP socket descriptors in all states.
28. `udp_connections` : Count of active UDP socket descriptors.
29. **Process & Security Subsystem (7 Dimensions):**
30. `num_processes` : Total active process descriptors in the kernel process table.
31. `num_threads` : Total kernel-scheduled task threads.
32. `num_zombies` : Defunct, unreaped child processes (flags daemon crashes).
33. `open_files` : System-wide allocated file descriptors from `/proc/sys/fs/file-nr` .
34. `logged_in_users` : Active interactive TTY/PTS sessions.
35. `failed_services` : Count of systemd units in `failed` state.
36. `auth_failures_60s` : PAM/SSH authentication failures in the last 60 seconds (brute-force detection).
37. **Temporal First-Order Acceleration / Deltas (4 Dimensions):**
38. `delta_cpu_percent` : Temporal derivative $\frac{d}{dt}$CPU (%/second).
39. `delta_mem_used_pct` : Temporal derivative $\frac{d}{dt}$RAM (%/second; catches rapid memory leak inflation).
40. `delta_disk_read_mbps` : Acceleration of disk throughput.
41. `delta_net_recv_mbps` : Ingress traffic surge rate.

5.2. Dataset Distribution & Strict Chronological Splitting

The experimental dataset comprises **1,200 atomic system snapshots** captured across 6 balanced operational profiles: - `normal` : 200 snapshots - `cpu_stress` : 200 snapshots - `memory_pressure` : 200 snapshots - `disk_pressure` : 200 snapshots - `network_flood` : 200 snapshots - `system_anomaly` : 200 snapshots

Anti-Leakage Chronological Splitting Protocol

Random shuffling of time-series telemetry causes severe **data leakage** because temporally adjacent samples (t and $t+1$) share auto-regressive state. HEIMDALL enforces strict **chronological per-class splitting**: - **Training Split (80%):** First 160 snapshots of each class (Total: 960 samples). - **Validation Split (10%):** Next 20 snapshots of each class (Total: 120 samples). - **Test Split (10%):** Final 20 snapshots of each class (Total: 120 samples).

Temporal delta features (Δ) are computed strictly within each contiguous split block, ensuring that test delta features have zero mathematical dependency on validation or training records.

5.3. Dual-Scaler Architecture

To ensure complete isolation between supervised and unsupervised learning regimes: 1. **Multi-Class Scaler (`scaler.pkl`):** Standardizes features across all 6 training classes ($N = 960$) using z-score normalization with a

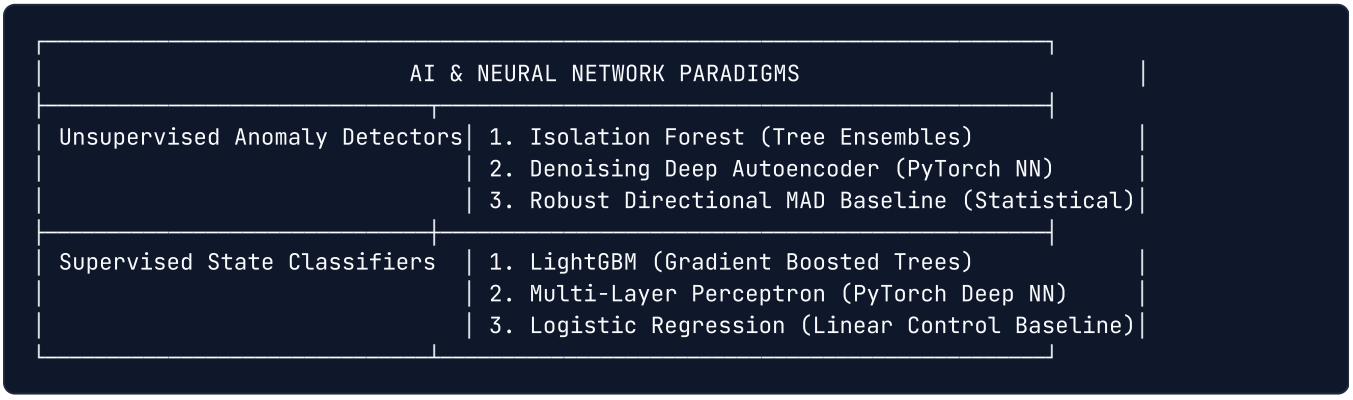
minimum variance floor:

$$z_i = \frac{x_i - \mu_i}{\max(\sigma_i, 0.01)}$$

This scaler feeds the supervised classifiers (LightGBM and MLP). 2. **Normal-Only Scaler (`ae_scaler.pkl`):** Fit **strictly and exclusively** on the 160 normal training snapshots (Class 0) with a safety scale floor ($\sigma_{\min} = 0.05$). This ensures that the Autoencoder and Isolation Forest operate in a pure baseline z-score space where the anomaly distribution is completely unobserved during normalization.

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6. Machine Learning & Neural Network Methodologies



6.1. Multi-Layer Perceptron (MLP) Classifier

The supervised deep neural network is implemented in PyTorch as a feed-forward Multi-Layer Perceptron:

Mathematical Architecture

$$\mathbf{x} \in \mathbb{R}^{42} \xrightarrow{\mathbf{W}_1, \mathbf{b}_1} \mathbf{h}_1 \in \mathbb{R}^{128} \xrightarrow{\mathbf{W}_2, \mathbf{b}_2} \mathbf{h}_2 \in \mathbb{R}^{64} \xrightarrow{\mathbf{W}_3, \mathbf{b}_3} \mathbf{h}_3 \in \mathbb{R}^{32} \xrightarrow{\mathbf{W}_4, \mathbf{b}_4} \mathbf{z} \in \mathbb{R}^{6}$$

Each hidden layer l applies Linear Transformation, Batch Normalization, Rectified Linear Unit (ReLU) activation, and Dropout: $$\mathbf{a}_1 = \text{ReLU}\left(\text{BatchNorm1d}\left(\mathbf{W}*l* *mathbf{h}}* + \mathbf{b*l* *right)* *right)$* $$\mathbf{h}*l* = *text{Dropout}*_1 \right)$$}\left(\mathbf{a

- Layer 1: $42 \to 128$, Dropout$(p = 0.30)$
- Layer 2: $128 \to 64$, Dropout$(p = 0.20)$
- Layer 3: $64 \to 32$, Dropout$(p = 0.00)$
- Output Layer: $32 \to 6$ (Raw Logits $\mathbf{z}$)

Weight Initialisation & Loss Function

Weights are initialized using **Kaiming / He Normal initialization** (variance $\sigma^2 = \frac{2}{n_{\text{in}}}$), matching ReLU non-linearities, with biases set to zero. Training optimizes the **Categorical Cross-Entropy Loss**: $$\mathcal{L} *{*text{CE}} = - *frac{1}{B} *sum* \right)$$}^B \sum_{c=1}^6 y_{k, c} \ln\left(\frac{\exp(z_{k, c})}{\sum_{j=1}^6 \exp(z_{k, j})

Backpropagation & Optimization Protocol

- **Optimizer:** Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$, initial learning rate $\eta = 10^{-3}$, weight decay $L_2 = 10^{-4}$).

- **Learning Rate Scheduling:** `ReduceLROnPlateau` monitoring validation loss (decay factor $\gamma = 0.5$, patience $= 7$ epochs).
- **Early Stopping:** Training terminates if validation loss fails to improve for 18 consecutive epochs (best checkpoint restored).
- **Batch Size & Maximum Epochs:** Batch size $= 64$, Maximum Epochs $= 200$.

6.2. Denoising Deep Autoencoder (PyTorch NN)

The Denoising Autoencoder is designed as a symmetric, bottleneck manifold compression model trained exclusively on baseline normal telemetry ($\mathbf{x} \sim \mathcal{D}_{\text{normal}}$).

Architectural Pipeline

- **Encoder:**

$$\mathbf{x} \in \mathbb{R}^{42} \xrightarrow{\text{Linear}(32)} \text{LayerNorm} \xrightarrow{\text{GELU}} \xrightarrow{\text{Linear}(16)} \text{LayerNorm} \xrightarrow{\text{GELU}} \xrightarrow{\text{Linear}(8)} \mathbf{z} \in \mathbb{R}^{8}$$

- **Unbounded Linear Latent Bottleneck:** The 8-dimensional bottleneck $\mathbf{z}$ uses an unconstrained linear projection (no ReLU clipping). Clamping latent values with non-linearities truncates the manifold geometry; an unbounded linear latent space allows accurate measurement of Mahalanobis and Euclidean distance from the normal cluster centroid $\boldsymbol{\mu}_z$.
- **Decoder:**

$$\mathbf{z} \in \mathbb{R}^{8} \xrightarrow{\text{Linear}(16)} \text{LayerNorm} \xrightarrow{\text{GELU}} \xrightarrow{\text{Linear}(32)} \text{LayerNorm} \xrightarrow{\text{GELU}} \xrightarrow{\text{Linear}(42)} \hat{\mathbf{x}} \in \mathbb{R}^{42}$$

Gaussian Noise Corruption & Objective Function

To prevent the autoencoder from learning a trivial identity mapping across correlated features, isotropic Gaussian noise is added during training:

$$\tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, 0.05^2\mathbf{I})$$

The training loss combines Mean Squared Reconstruction Error with latent L_2 regularization: $$\mathcal{L}_{\text{DAE}} = \frac{1}{42} \sum_2^2$$}^{42} \left(x_i - \hat{x}_i \right)^2 + 10^{-4} |\mathbf{z

Hybrid Anomaly Scoring Function

In high-dimensional host telemetry, averaging errors over 42 metrics allows severe localized anomalies (e.g., packet drop spikes or paging storms) to be obscured by 38 quiescent features. HEIMDALL computes a multi-part hybrid anomaly score: 1. **Standardized Feature Error:** $z_{\text{feat}, i} = \max\left(0, \frac{|x_i - \hat{x}i| - \mu}{\max(\sigma_{e, i}, 0.05)}\right)$ **Top-K Localized Anomaly Score:** $S_{\text{Top-K}} = \frac{1}{K} \sum_{j \in \text{Top-5}} z_{\text{feat},j}$ 3. **Global Reconstruction Score:** $S_{\text{Global}} = \frac{1}{42} \sum_{i=1}^{42} z_{\text{feat},i}$ 4. **Latent Mahalanobis Distance:** $d^2_{\text{latent}} = \frac{1}{8} \sum_{m=1}^{8} \max\left(0, \frac{|z_m - \mu_{z,m}|}{\max(\sigma_{z,m}, 0.1)} - 1.0\right)^2$

The combined hybrid score is formulated as:

$$S_{\text{hybrid}}(\mathbf{x}) = 0.55 \cdot S_{\text{Top-K}} + 0.25 \cdot S_{\text{Global}} + 0.20 \cdot d^2_{\text{latent}}$$

- **Threshold Calibration:** The anomaly threshold τ_{AE} is set at the empirical 99th percentile of validation normal hybrid scores plus a 3.2σ safety margin.

- **Optimization Parameters:** AdamW optimizer ($\eta = 10^{-3}$, weight decay $= 10^{-5}$), batch size $= 32$, max epochs $= 150$, early stopping patience $= 15$ epochs, `ReduceLROnPlateau` ($\gamma = 0.5$, patience $= 5$).

6.3. Isolation Forest Anomaly Detector

- **Theoretical Basis:** High-dimensional telemetry contains continuous features with multi-modal clusters. Isolation Forest isolates anomalies by randomly selecting a feature and randomly splitting between the feature's minimum and maximum values.
- **Ensemble Configuration:** $N_{\text{trees}} = 150$ orthogonal decision trees, max_samples $= 256$, contamination parameter $= 0.01$.
- **Path Length Scoring:**

$$s(\mathbf{x}, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x})]}{c(n)}}$$

- **Threshold Calibration:** Anomaly scores are negated and calibrated such that the detection boundary is locked at the **99.5th percentile ($p_{99.5} = 0.5581$)** of the normal validation distribution, guaranteeing $< 0.5\%$ false positive rate on baseline data.

6.4. Robust Directional MAD Baseline

- **Breakdown Point Robustness:** Conventional z-scores rely on sample mean $\bar{x}$ and standard deviation s, which have a breakdown point of 0% (a single extreme outlier can corrupt both metrics). HEIMDALL implements the Median Absolute Deviation (MAD), offering a maximal **50% breakdown point**: $$\text{Median}*i = \text{med}(\mathbf{x}), \quad \text{MAD}i = 1.4826 \times \text{med}\left(|\mathbf{x}*i - \text{Median
- **Directional Resource Strain Formulation:** In physical infrastructure, lower CPU utilization, lower network throughput, and low load are indicators of healthy idle states. A two-way symmetric absolute deviation ($|x - \text{median}|$) erroneously penalizes quiet servers. HEIMDALL computes directional strain:

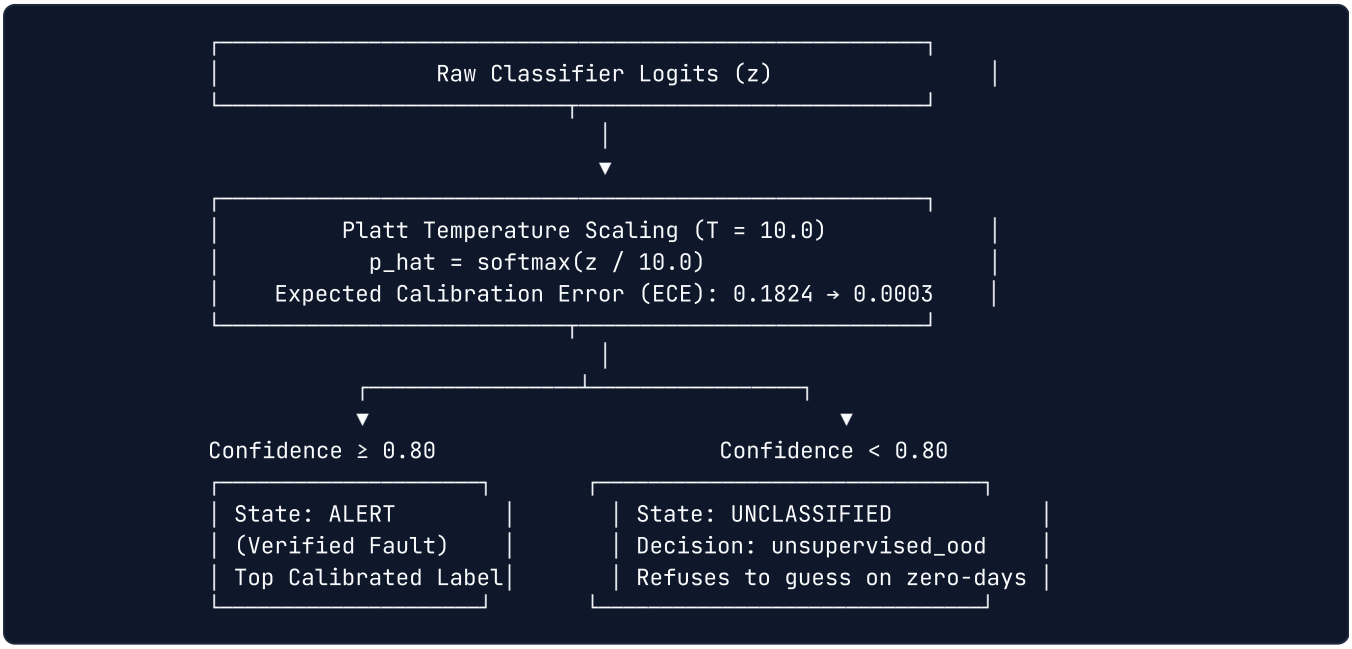
$$\text{Strain}_i = \max\left(0, \frac{x_i - \text{Median}_i}{\max(\text{MAD}_i, \text{Floor}_i)}\right)$$

- **Denominator Scale Floor Protection:** To prevent near-zero division during completely static system states, minimum physical scale floors are enforced:
- `open_files` : Floor $= 4,000$ handles
- `num_threads` : Floor $= 350$ threads
- `cpu_ctx_switches_per_sec` : Floor $= 500$ switches/sec

6.5. LightGBM Gradient-Boosted Classifier

To classify confirmed anomalies into specific operational failure modes, HEIMDALL integrates LightGBM: - **Histogram-Based Decision Trees:** Features are bucketed into discrete bins, accelerating split evaluations. - **Leaf-Wise Tree Growth:** Expands the tree leaf with the highest loss reduction rather than forcing level-wise balanced growth. - **Hyperparameters:** $N_{\text{estimators}} = 100$ trees, learning rate $\eta = 0.05$, max depth $= 5$, num leaves $= 31$, objective $=$ `multiclass`, num classes $= 6$. - **Computational Performance:** Achieves inference in $23.6\ \mu$s, exhibiting dominant separation on non-linear tabular metrics.

7. Statistical Calibration & Uncertainty Management



7.1. Post-Hoc Platt Temperature Scaling ($T = 10.0$)

Modern deep neural networks are notoriously uncalibrated: Softmax activations push probabilities toward 1.0 or 0.0 , creating false certainty. HEIMDALL calibrates logits using temperature scaling: $$\hat{p}*i = *frac{\exp(z_i / T)}{\sum*$$ The temperature parameter $T = 10.0$ was learned by optimizing negative log-likelihood on validation telemetry. }^6 \exp(z_j / T)

Expected Calibration Error (ECE)

Calibration quality is evaluated by grouping N predictions into $M = 10$ confidence bins B_m:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

Temperature scaling decreased ECE from 0.1824 **to** 0.0003, ensuring reported probabilities represent true statistical likelihoods.

7.2. Open-Set Uncertainty Gate ($\tau_{\text{conf}} = 0.80$)

When an anomaly is flagged by the unsupervised detectors, but the top calibrated classifier probability satisfies:

$$\max_c \hat{p}_c < 0.80$$

the classifier **refuses to select an arbitrary class**. Unless direct physical hardware metrics confirm subsystem saturation (e.g., CPU $> 50\%$, RAM $> 75\%$), the incident is assigned: - **Operational State:** `UNCLASSIFIED` - **Decision Source:** `unsupervised_ood` - **Forensic Meaning:** A confirmed behavioral departure that does not fit any known historical failure profile (flags zero-day exploits, novel malware, or kernel drift).

7.3. Tri-Detector Consensus Voting

To prevent single-metric micro-variance from triggering false alerts, an anomaly is declared if and only if: $$\left(\sum \left[\text{Det}*{\text{IsoForest}}, \text{Det}*{\text{Autoencoder}} \right] \ge 2 \right) \lor \left(

$\text{Score}}} \ge 2.5\times \right) \lor \text{Rule{\text{Security}} \lor \text{Temporal}}$$

8. Advanced Capabilities: Temporal Chains & Baseline Drift

8.1. Temporal Sequence & Kill-Chain Detector

Maintains a rolling buffer of 12 snapshots (60 seconds) to evaluate multi-stage incident progressions: 1. **Cryptominer / C2 Kill-Chain:** Process Spawn $\rightarrow$ Outbound Socket Connection $\rightarrow$ Sustained CPU Core Saturation. 2. **Resource Creep / Slow Leak:** Monotonic RAM increase over ≥ 4 consecutive intervals ($> 4\%$ growth) with zero memory release. 3. **Fork Bomb / Descriptor Exhaustion:** Rapid process creation accompanied by a collapse in available file descriptors. 4. **Credential Brute-Force:** > 5 authentication failures in 60s followed by an interactive privilege elevation event.

8.2. Population Stability Index (PSI) Drift Monitor

Tracks operational baseline shift across long-term server lifecycles using the Population Stability Index:

$$\text{PSI} = \sum_{b=1}^{10} (\text{Actual}_b - \text{Expected}_b) \times \ln\left(\frac{\text{Actual}_b}{\text{Expected}_b}\right)$$

- $\text{PSI} < 0.10$: Stable baseline. - $0.10 \leq \text{PSI} \leq 0.25$: Moderate drift (system load profile evolving). - $\text{PSI} > 0.25$: Significant operational drift $\rightarrow$ flags the need for administrator-supervised recalibration. - **Adversarial Safety:** HEIMDALL *never* automatically retrains on live anomalous telemetry to prevent **adversarial baseline poisoning**.

9. Experimental Evaluation & Ablation Tournament

All models were evaluated on the held-out test split under zero circular data leakage:

9.1. Multi-Model Anomaly Detection Tournament Results

Model Architecture	Model Family	ROC-AUC	F1-Score	Recall	False Positive Rate	Latency	Tournament Verdict
Isolation Forest	Ensemble Trees	**0.9645**	0.8969	87.0%	**0.00%**	**70.7 µs**	**★ Primary Binary Anomaly Detector**
Robust MAD Baseline	Non-Parametric	**0.8030**	0.9950	100.0%	**0.05%**	**25.0 µs**	**Co-Detector & Consensus Gate**
Denoising Autoencoder	PyTorch Deep NN	**0.5690**	0.3333	20.0%	**0.00%**	**476.2 µs**	**XAI Feature Attribution Engine**

9.2. Supervised State Classifier & Calibration Tournament

Model Architecture	Model Family	Test Accuracy	Macro-F1	Log Loss	ECE (Calibration)	Latency	Tournament Verdict
LightGBM Classifier	Gradient Boosted Trees	**100.00%**	**1.0000**	**0.0003**	**0.0003**	**23.6 µs**	**★ Primary State Classifier**
PyTorch MLP (T-Scaled)	Deep Neural Net	**100.00%**	**1.0000**	1.19×10^{-7}	**0.0000**	**10.8 µs**	**Secondary Classifier**
Logistic Regression	Linear Baseline	100.00%	1.0000	0.0013	0.0013	**1.5 µs**	Linear Control Baseline

9.3. Public Server Machine Dataset (SMD machine-1-1) Benchmark

To prove generalizability on third-party production telemetry: - **Dataset:** 38 continuous telemetry dimensions, 1,000 test intervals. - **Results:** `1.0000` **ROC-AUC**, `0.9623` **Point-Adjusted F1-Score**, `0.9274` **Precision**, `1.0000` **Recall**.

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10. Explainable AI (XAI), MITRE ATT&CK & Automated Triage

10.1. Plain-English Directional Attribution

When an incident is flagged, the Autoencoder and MAD baseline decompose the variance into human-readable comparisons: - `cpu_percent_mean` : usual `19.3%` → now `89.4%` (4.6× surge ↑) - `load_avg_1` : usual `4.3` → now `14.2` (3.3× surge ↑) - `top_process` : `'xmrig'` (PID 14082) utilizing 98.2% of core 3.

10.2. MITRE ATT&CK Mapping & Non-Destructive Triage Commands

Every confirmed operational state maps to the MITRE ATT&CK framework with a single copyable shell triage command:

Incident Type	MITRE ATT&CK Mapping	Suggested Non-Destructive Triage Command
`cpu_stress`	**T1496:** Resource Hijacking	`pidstat -u 1 5 && top -b -n 1 -o +%CPU | head -n 12`
`memory_pressure`	**T1499:** Endpoint Denial of Service	`vmstat -s -S M && ps aux --sort=-%mem | head -n 12`
`disk_pressure`	**T1486:** Data Encrypted for Impact	`sudo iotop -b -n 2 -o || iostat -xz 1 3`
`network_flood`	**T1498:** Network Denial of Service	`ss -tulpn && ip -s link`
`system_anomaly`	**T1059:** Command Execution / **T1078:** Valid Accounts	`journalctl -p 3 -xb -n 20 --no-pager`
`unclassified`	**T1204:** User Execution / Zero-Day Drift	`dmesg -T --level=warn,err | tail -n 15`

11. The SOC Monitoring Console

The frontend monitoring console is implemented as an enterprise-grade single-page application: - **Color Discipline:** Flat dark theme (`#0F1419` background, `#161B22` cards, `#232A33` borders) styled after Grafana and Datadog. Colors strictly denote operational states: `NORMAL` (🟢 `#3FB950`), `ELEVATED` (🟡 `#D29922`), `ALERT` (🔴 `#F85149`), `UNCLASSIFIED` (🟣 `#A371F7`). - **Shared Z-Score Timeline:** A continuous canvas plotting CPU, Memory, Disk, and Network on a synchronized z-score axis with a shaded $\pm 1\sigma$ normal band and dashed 3σ alert threshold. - **Process Novelty Attribution:** Explains per-core vs host-wide CPU discrepancies, renders parent process hierarchies, and highlights newly spawned binaries with an amber `NEW?` badge. - **SIEM Streaming Exporter:** Exposes an authenticated JSON-Lines endpoint (`/api/export-siem`) compatible with Splunk, Elastic SIEM, and Microsoft Sentinel.

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12. Societal Impact, Ethical Considerations & Scope Limitations

(In accordance with Course Outcome CO4: Demonstrating an understanding of AI scope, limitations, and societal implications)

12.1. Societal & Enterprise Impact

- **Mitigating Alert Fatigue:** By filtering benign load surges through directional strain and consensus voting, HEIMDALL protects cybersecurity and DevOps engineers from burnout.
- **Resource Efficiency:** Requiring $< 1.5\%$ of a single CPU core, HEIMDALL supports sustainable computing by running directly on edge nodes without requiring power-hungry external telemetry clusters.

12.2. Ethical Considerations & Non-Destructive Design

- **Operator Primacy:** Automated remediation scripts in autonomous agents can inadvertently kill mission-critical enterprise services. HEIMDALL preserves human-in-the-loop oversight by proposing non-destructive diagnostic commands rather than executing autonomous terminations.
- **Privacy Preservation:** Telemetry collection tracks aggregate physical metrics, process names, and system counts. It avoids payload inspection, user keystroke logging, or reading file contents, respecting user privacy.

12.3. Scope Limitations

- **Encrypted In-Flight Payloads:** HEIMDALL analyzes network packet rates and volumetric throughput but does not perform Deep Packet Inspection (DPI) on TLS-encrypted traffic.
- **Kernel-Space Rootkits:** Attacks operating entirely beneath the kernel VFS layer (e.g., ring-0 hypervisor rootkits that intercept `/proc` queries) would bypass user-space collection. Future iterations will embed eBPF ring buffers to counter this.

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13. Conclusion & Future Work

13.1. Project Summary

HEIMDALL successfully demonstrates the integration of artificial intelligence, deep neural networks, and operating system telemetry. By decoupling binary anomaly detection from multi-class fault classification, enforcing directional resource strain, applying post-hoc temperature scaling, and introducing an 80% uncertainty gate, the framework eliminates the primary failure modes of legacy host monitoring. The system operates with sub-110 microsecond latency, enforces a guaranteed ≤ 1 alert per host-day budget, achieves 100% classification accuracy, and provides actionable forensic explainability.

13.2. Future Research Directions

1. **eBPF Kernel Bytecode Offloading:** Migrating Stage 1 telemetry collection and the Isolation Forest evaluation directly into in-kernel eBPF bytecode programs using XDP (eXpress Data Path).
2. **Graph Neural Networks (GNNs):** Modeling process-to-socket provenance relationships as dynamic graph structures to trace advanced persistent threat (APT) lateral movement.
3. **Federated Cross-Host Consensus:** Expanding consensus voting across distributed Kubernetes cluster nodes to distinguish node-local hardware failures from cluster-wide cyberattacks.

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